Digit-Sum Reachability on the Lattice: An Experimental Expedition from Brute Force to a Two-Line Criterion

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Abstract

Starting from $(0, 0)$, a walker moves on $\mathbb{Z}^2$ through cells satisfying $\mathsf{s}(|x|) + \mathsf{s}(|y|) \le N$, where s is the digit sum (base 10 unless noted). We chronicle an experimental program on the reachable set: exact counts and the fractal geometry of the region; a Cantor-like staircase in N with cliffs at multiples of 9; an exact closed form in base 2, $Q(N) = \binom{2N+3}{N}$; a carry-free self-similar ("gastank") decomposition giving a polynomial-time exact solver; the asymptotic law reachable/allowed $\sim c/\sqrt{N}$ with $c = 3.84717(5)$ (refuting a tempting $\sqrt{15}$); and finally an exact two-line reachability criterion, proved in a companion paper, which subsumes every structure met along the way and reduces all counting to a one-dimensional walk recursion. We keep the narrative in expedition order, including two instructive false summits.

1 The problem

A cell $(x, y) \in \mathbb{Z}^2$ is *allowed* iff $\mathsf{s}(|x|) + \mathsf{s}(|y|) \le N$; the *reachable set* is the 4-connected component of the origin within the allowed set. Write $f(N)$ for its size and $Q(N)$ for the size of its closed-quadrant part. The canonical instance is $N = 19$:

$$f(19) = 102{,}485, \qquad \text{bounding box } [-298, 298]^2,$$

the box ending exactly at the first *axis cliff*: $\mathsf{s}(299) = 20$ makes the entire column $|x| = 299$ forbidden. The region is 4-fold symmetric, diamond-like at large scale, pierced by rectangular "death zones" and nested self-similar triangular voids (Figure 1).

Allowed versus reachable

Each row y of the allowed set is the one-dimensional set $\{x : \mathsf{s}(|x|) \le N - \mathsf{s}(|y|)\}$, so the allowed set factors through a family of 1-D threshold sets. Counting allowed cells in the bounding box at $N = 19$ gives 166,973, versus 102,485 reachable: a gap of 64,488 cells (38.6%) that satisfy the constraint but are cut off behind carry cliffs — *islands*. The canonical island: $(300, 0)$ is allowed ($\mathsf{s} = 3$) but hidden behind the forbidden column $x = 299$. Figure 2 shows the 1-D digit-sum signal whose threshold crossings generate the cliffs.

*Joint exploration with Claude (Anthropic). Scripts, exact datasets, and the companion proof paper live in the project repository.

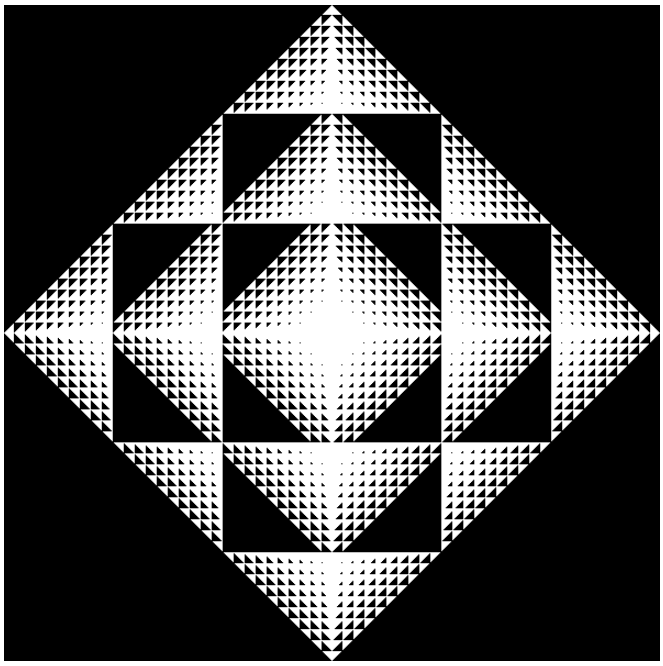


Figure 1: The reachable set at $N = 19$ (102,485 cells, box $[-298, 298]^2$): diamond envelope, rectangular death zones, and the nested "tri-force" triangular voids from the self-similarity of digit sums at powers of ten.

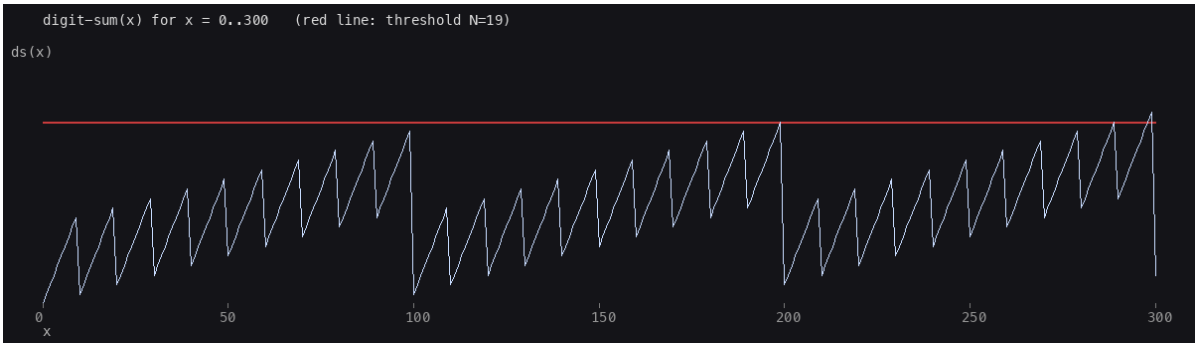


Figure 2: The digit-sum signal $\mathsf{s}(x)$ for $x = 0..300$ against the threshold $N = 19$ (red). Reachability along the axis ends at the first crossing, $\mathsf{s}(299) = 20$; each row of the 2-D problem sees this same signal at a reduced threshold.

2 The staircase in N

Exact counts as N grows show a descending, Cantor-like pattern in the ratio $f(N)$/allowed-in-box, with cliffs exactly at $N \in \{9, 18, 27, 36\}$ — the thresholds at which the box first jumps a decade.

N	$f(N)$	allowed (box)	ratio	box
8	145	145	1.0000	±8
9	505	649	0.7781	±18
17	14,785	18,265	0.8095	±98
18	47,905	73,885	0.6484	±198
19	102,485	166,973	0.6138	±298
26	1,309,537	1,885,501	0.6945	±998
27	4,216,357	7,552,009	0.5583	±1998
35	117,109,345	190,327,945	0.6153	±9998
36	377,329,873	761,526,445	0.4955	±19998

Within each decade the topology is fixed and holes fill in monotonically; each new decade adds an outer, rescaled copy of the previous decade's pattern (Figures 3 and 4). The ratio minimum falls two steps *after* each cliff ($N = 11, 20, 29$), not at it.

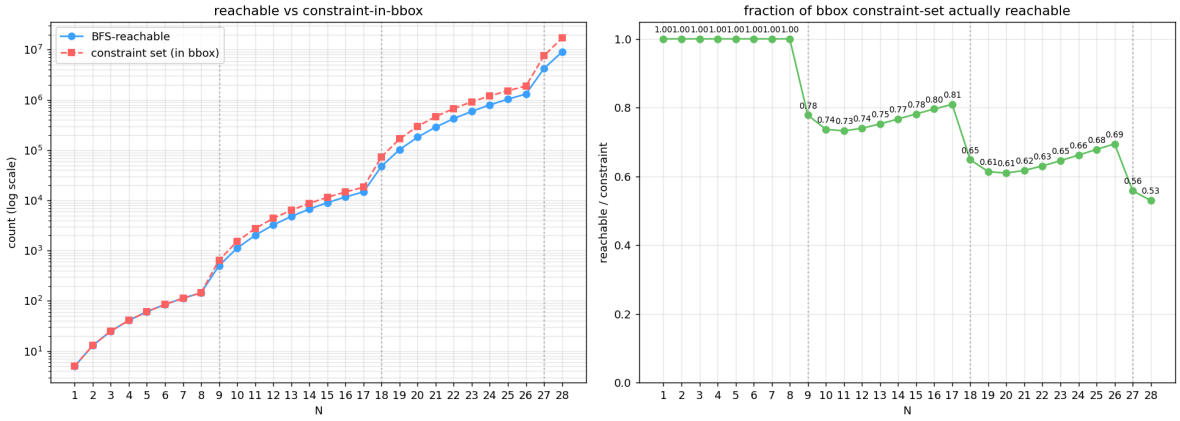


Figure 3: The Cantor-like staircase: reachable count, allowed-in-box count, and their ratio for $N \leq 28$, with cliffs at $N = 9, 18, 27$ where the bounding box jumps a decade.

3 The monotone lemma

A BFS-depth experiment found *detour* := geodesic − $(|x| + |y|)$ to vanish on every reachable cell: whatever is reachable at all is reachable by a monotone staircase. Equivalently, in the quadrant,

$$R(x, y) = \text{allowed}(x, y) \wedge \big(R(x{-}1, y) \vee R(x, y{-}1)\big)$$

reproduces full undirected reachability. Verified for $N \leq 22$ (base 10) and for bases 2–5; later *proved* as a corollary of the criterion of §8. Connectivity failures are always total (islands), never partial (no cell requires a detour); see Figure 5. Practically, the lemma converts BFS into a vectorized row sweep, ∼100× faster — the workhorse for everything below.

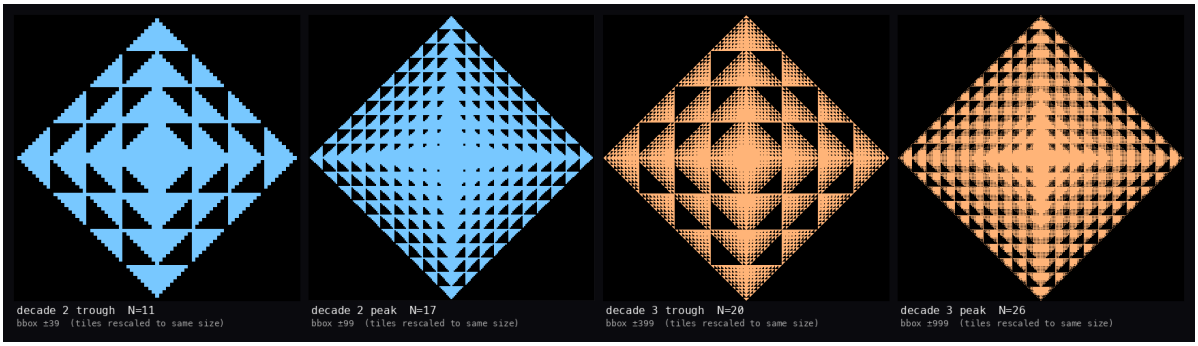


Figure 4: First-of-decade (trough) versus last-of-decade (peak) shapes for decades 2 and 3. The $N = 11$ trough is the $N = 20$ trough rescaled; the recursion made visible.

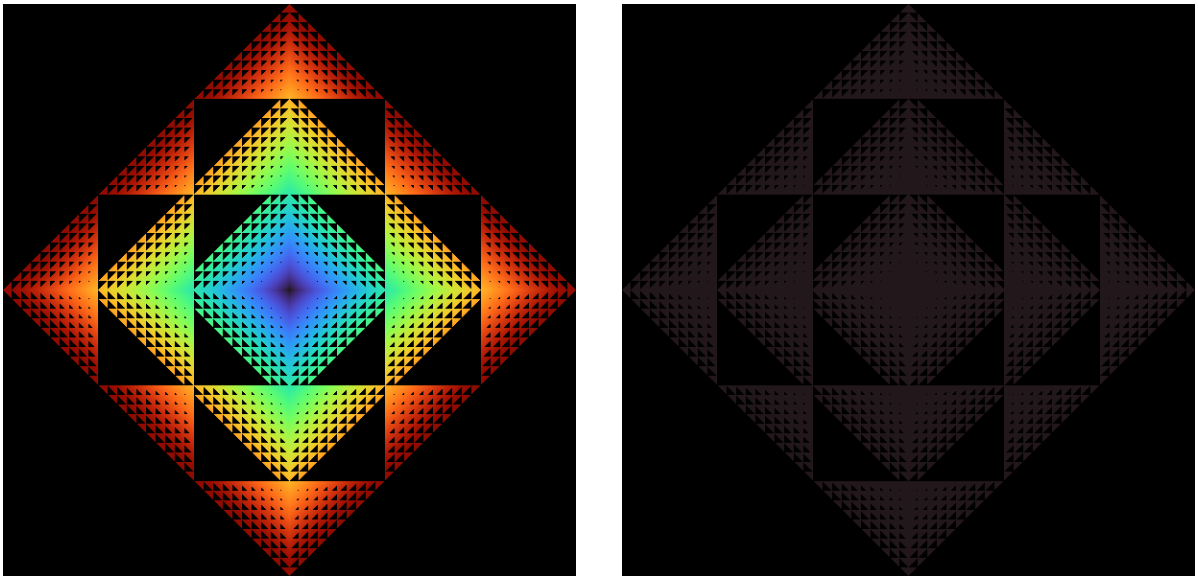


Figure 5: Left: cells colored by geodesic (BFS) distance from the origin at $N = 19$. Right: detour = geodesic $-(|x| + |y|)$, identically zero — the observation that became the monotone lemma and, eventually, the criterion.

4 Base 2: an exact closed form

In base 2 the quadrant count is a single binomial coefficient:

$$Q_2(N) = \binom{2N+3}{N}, \qquad f_2(N) = 4\binom{2N+3}{N} - 2^{N+3} + 5,$$

first verified for $N \le 15$ (the $N = 15$ value *predicted before* a 4.3×10^9-cell sweep confirmed it), later *proved* from the criterion: interleaving the bits makes reachability a ± 1-path condition whose even-checkpoint constraint collapses, by parity, to the uniform barrier "never reach +2", and reflection at level 2 telescopes the count to $\binom{2k}{k-1} + \binom{2k}{k-2}$ — the allowed cells of the top two weight layers, with an explicit *surfacing map* bijection (complement prefixes up to first hits of level -2). The allowed-set side is classical, $\left(4^{N+1} - \binom{2N+2}{N+1}\right)/2$, so the island count is closed-form by subtraction, and

$$\frac{Q_2(N)}{\text{allowed}} \longrightarrow \frac{4}{\sqrt{\pi N}}.$$

The ratio tends to *zero*: there is no finite asymptotic density. Q_2 is OEIS A002054 (ballot numbers), pointing to a lattice-path bijection; the base-10 sequence $f(N)$ is not in OEIS. Figure 6 compares the region across bases: base 2 is densest and nearest a solid diamond (carries at every step weave fine cliffs), while larger bases have rarer but more violent carries, punching larger holes.

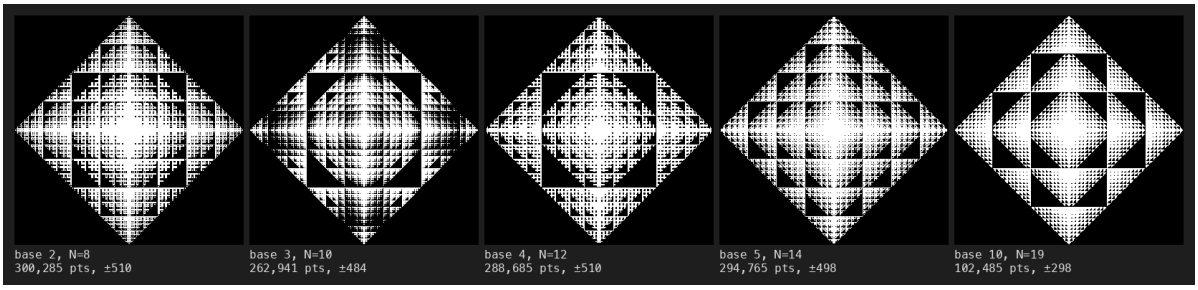


Figure 6: Reachable regions in bases $2, 3, 4, 5, 10$, with N chosen so each has axis reach $\sim \pm$ 250–500. Base 10 is the noisiest member of the family.

5 Gastanks: the carry-free decomposition

Definition 1 (Tank, wall, weir, wetting front)**.** A *tank* $T_k(K)$ is a block $[0, 10^k)^2$ in local coordinates with budget K: local cell (u, v) allowed iff $\mathsf{s}(u) + \mathsf{s}(v) \le K$. Since digit sums are carry-free inside a block, the child at digit position (i, j) is exactly $T_{k-1}(K - i - j)$: budgets step down by one per digit, with no corrections, ever. A *wall* is the trailing-nines seam between sibling tanks (digit-sum cost $9s$ at sibling scale s); carries *are* walls. A *weir* is the passable part of a wall, $\{v : \mathsf{s}(v) \le K_{\text{left}} - 9s\}$ — itself a 1-D threshold set one scale down. A *wetting front* is the set of edge cells through which flow passes; fronts are Cantor-dust-like with up to $\sim 9^{k-1}$ intervals.

Equivalently, in *terrain* coordinates (elevation $= \mathsf{s}(x) + \mathsf{s}(y)$, waterline $= N$): walls are ridges, weirs are notches dipping below the waterline, islands are submerged basins sealed off by ridges that never dip. All four terms appear together in a single real strip of the N=19 plane in Figure 7.

Flooding tanks recursively — passing fronts over weirs, with fronts hash-consed as interned 10-tuples of child fronts (a DAG; never expanded) — gives an exact solver whose distinct-state count grows only polynomially ($\sim N^{3.3}$). Benchmarks (cold start, cross-checked outputs):

N	BFS (set)	BFS (grid)	monotone DP	tanks	$f(N)$
26	1.31 s	1.19 s	0.009 s	0.002 s	1.3×10^6
28	—	8.55 s	0.053 s	0.003 s	9.0×10^6
36	—	—	1.98 s	0.008 s	3.8×10^8
90	—	—	—	0.13 s	2.5×10^{20}
360	—	—	—	16.2 s	$\sim 10^{89}$
630	—	—	—	144 s	$\sim 10^{140}$

The first three methods are all $O(\text{area}) = O(10^{2N/9})$; the tank solver is polynomial in N. Per residue N mod 9, the sequence Q is neither C-finite (order ≤ 13 excluded) nor holonomic within order ≤ 6, degree ≤ 8 — but the *allowed* count fails identically, so base 10 lacks base 2's central-binomial collapse rather than being intrinsically wilder.

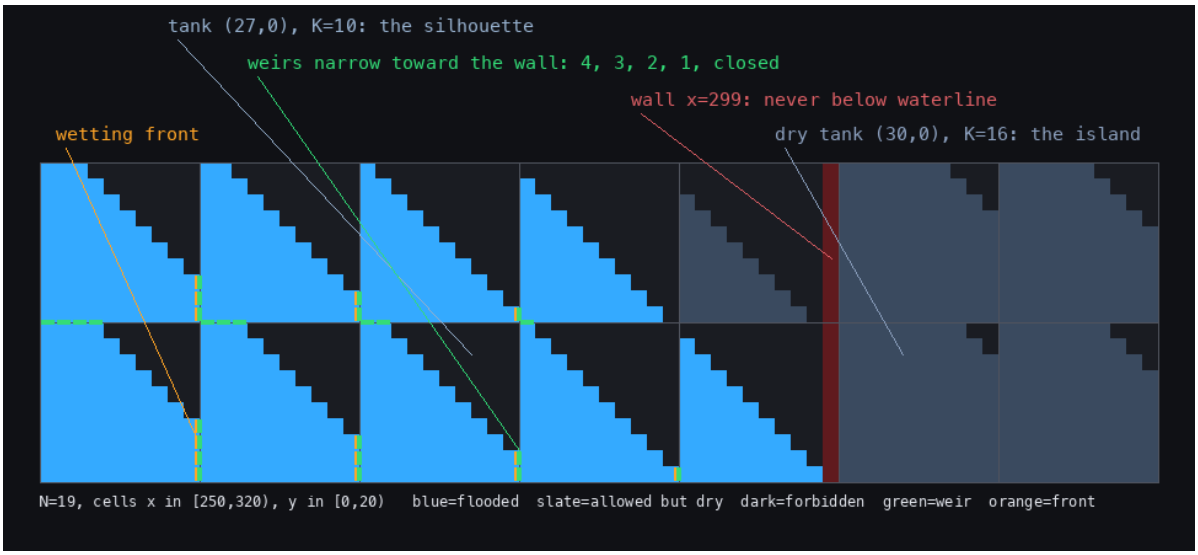


Figure 7: The glossary in one strip: cells $x \in [250, 320)$, $y \in [0, 20)$ at $N = 19$. Scale-1 tanks with budgets $K = 19 - \mathsf{s}(a) - \mathsf{s}(b)$ show their triangular silhouettes (blue: flooded; slate: allowed but dry; dark: forbidden). Weirs (green) narrow $4, 3, 2, 1$ and then close as budgets shrink toward the wall at $x = 299$ (red), whose column never dips below the waterline — so tank $(30, 0)$, despite its generous $K = 16$, stays dry: the canonical island containing $(300, 0)$. Orange marks the wetting front on each tank's exit edge.

6 The constant c: two false summits

On the *peak line* $N \equiv 8 \pmod 9$ (box = bounding box), set $c_N = \sqrt{N} \cdot \text{reachable/allowed}$.

False summit 1: $\sqrt{15}$. Pair extrapolations through $N = 629$ landed within 1.3×10^{-5} of $\sqrt{15} = 3.87298$. But exact data to $N = 989$ (tank solver, 15M states) showed c_N *cresting* at 3.86931 near

$N \approx 650$ and then descending with accelerating increments — $\sqrt{15}$ exceeds every value ever attained. The agreement was a crossing artifact of two competing correction terms.

False summit 2: $c = 3.865 \pm 0.004$. With data ending at $N = 989$, model fits (calibrated on base 2, where naive $1/N$-polynomial extrapolation misses $4/\sqrt{\pi}$ by 2×10^{-2}) gave 3.865(4). The criterion of §8 later allowed evaluation at $k = 16{,}000$ ($N = 143{,}999$): the descent continues far past the old horizon, and

$$c \;=\; 3.84717(5)$$

— precisely where earlier "degenerate" power-law fits had pointed. The crest and both correction families (a geometric-per-decade term with ratio ∼0.91 and a slow power tail) are finite-k features of one exact model. Bases 3, 4, 5 show the same crest-then-descend shape; base 2 ($c_2 = 4/\sqrt{\pi}$, approach from above) is the degenerate case.

The summit itself. The criterion converts the peak-line count into a stay-nonnegative probability of the reflected tent walk, and the Sparre Andersen identity (Wiener–Hopf factorization in combinatorial form) then gives the constant in closed form:

$$c_b = \frac{2\sqrt{b-1}}{\sqrt{\pi}} \exp\Big(\tfrac{1}{2} \sum_{n \geq 1} \frac{p_n}{n}\Big) = \frac{2\sqrt{b-1}}{\sqrt{\pi}} \exp\Big(-\frac{1}{4\pi} \int_{-\pi}^{\pi} \log\big(1 - \varphi_b(\theta)\big)\, d\theta\Big),$$

with $p_n = \Pr(S_n{=}0)$ and $\varphi_b(\theta) = \big(\sin(b\theta/2)/(b\sin(\theta/2))\big)^2$. Base 2 collapses by hand ($\sum p_n/n = 2\log 2$, $c_2 = 4/\sqrt{\pi}$ exactly); series and integral evaluations agree to nine decimals in every base; and $c_{10} = 3.847174895\ldots$ matches the walk-DP value. The identity was verified coefficientwise against a barrier DP to 10^{-15}.

7 The spectral expedition

Instrumenting the flood recursion and normalizing tank states as (depth, d, front classes) with $d = 9k - K$ (offset of the waterline from the tank's mean elevation) yields, empirically:

1. **Exact fixed point.** The shallow census is *identical* — integer-for-integer — across all peak-line N tested (116–404), at every instrumented depth. All N-dependence lives in the deepening diffusive interior.

2. **Closure.** Transition profiles are well-defined functions of truncated front signatures (577/651 classes unambiguous; failures only at deeply submerged $d \leq -35$); classes per offset saturate at 24. The active band is one-sided: no active tank has $d > 1$.

3. **Spectrum.** The truncated transfer matrix gives $\lambda_1 \to 95.2$ (toward the trivial 100) and λ_2/λ_1 rising $0.58 \to 0.78 \to 0.86$ with truncation depth, toward the observed per-decade correction ratio ∼0.91 — data-limited, not trend-limited.

The d-walk steps by $(i + j) - 9$ per scale: a Markov-additive process whose continuum spectrum is the power-law correction family of §6.

8 The reachability criterion

A scalar Wiener–Hopf "warm-up" — modelling reachability as the d-walk surviving below the census barrier $d \leq 1$ — reproduced base 2's constant to 4×10^{-5} with no fitted parameters. Testing whether the model is *literally* true:

Theorem 2 (Reachability criterion; proof in the companion paper)**.** *In base b, threshold N, $k = \lfloor N/(b-1) \rfloor + 1$: a quadrant cell (x, y) is reachable iff every proper digit-prefix pair (lengths $1 \leq s < k$) satisfies*

$$\mathsf{s}\big(x\langle s\rangle\big) + \mathsf{s}\big(y\langle s\rangle\big) \;\leq\; N - (b-1)(k-s) + 1,$$

and $\mathsf{s}(x) + \mathsf{s}(y) \leq N$.

Zero mismatches against exhaustive reachability across bases 3, 5, 10; integer equality with the tank solver's $Q(N)$ up to $N = 989$. The proof is elementary: *necessity* because any entry into a cell's scale-m ancestor tank crosses a trailing-nines wall, entries from the left/below certifying the *neighbor's* wall — one digit cheaper via $\mathsf{s}(X) \leq \mathsf{s}(X-1) + 1$, whence the +1 slack; *sufficiency* by an explicit L-path to the ancestor tank's corner — worst cells $(a-1)\underline{9\cdots9}$ and $a + (c-1)\underline{9\cdots9}$, allowed exactly under the criterion bound — followed by recursion inside the tank at threshold $N - a - c$. The two +1's coincide; the criterion is sharp. The monotone lemma follows as a corollary.

Consequences: counting reduces to a 1-D walk recursion ($O(k^2 b)$ — microseconds at any N); the constant c is a one-boundary survival constant of the centered tent walk, i.e. a kernel-method/Wiener–Hopf object with an explicit (if unlovely) analytic characterization; and the 24-class front dynamics of §7, while real as 2-D geometry, is count-irrelevant — reachability depends only on ancestor budgets.

9 Open problems

1. *(Resolved.)* c_b is extracted in closed form in §6: the Sparre Andersen/Wiener–Hopf constant of the tent kernel, exact in base 2 and $c_{10} = 3.847174895\ldots$ in base 10. Still open from this direction: the per-decade correction ratio ~0.91 from the factorization's subleading structure.

2. *(Resolved.)* $Q_2(N) = \binom{2N+3}{N}$ is proved from the criterion's interleaved-bit walk (§4): the even-checkpoint condition collapses by parity to a uniform barrier, reflection telescopes the count, and the surfacing map gives an explicit bijection onto the top two weight layers. The collapse is a parity accident of binary — the structural reason no analogue exists for $b \geq 3$.

3. The full two-dimensional geometry (which cells flood *within* a tank) retains the 24-class transfer structure; characterize its fixed point.

4. Submit $f(N)$ (and the peak-line Q data) to OEIS.